Advanced Resistance Training Strategies for Bodybuilding: Tools for Muscle Hypertrophy

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ABSTRACT

Maximizing muscle hypertrophy is a primary goal for bodybuilders and resistance-trained individuals, driving the exploration of advanced resistance training strategies that extend beyond traditional programming. This review comprehensively examines the theories, mechanisms, and practical applications of various advanced training techniques, including supersets, drop sets, lengthened partials, pre-exhaustion training, rest-pause training, forced repetitions, pyramidal training, interset stretching, and blood flow restriction. These methods aim to enhance mechanical tension, metabolic stress, and muscle activation in effort to promote muscle growth. Although many of these strategies offer time-efficient alternatives to conventional training, current evidence suggests that they do not consistently surpass traditional resistance training in muscle growth when volume is equated. Despite their widespread use in bodybuilding, research is often limited by short study durations, small sample sizes, and a lack of studies involving competitive bodybuilders. This review highlights key findings from existing literature, identifies gaps requiring further investigation, and provides practical recommendations for integrating advanced training strategies into evidence-based hypertrophy programs.

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INTRODUCTION

In the pursuit of maximizing muscle growth, bodybuilders have long sought to optimize their training methods, often turning to advanced resistance training strategies in effort to push the boundaries of conventional exercise programming (42). These advanced and/or time-saving techniques, such as supersets, drop sets, lengthened partials, pre-exhaustion training, rest-pause training, forced repetitions, pyramidal training, interset stretching, and blood flow restriction (BFR), have gained widespread popularity for their ability to significantly boost training density and/or intensity. In addition, the use of time-efficient resistance training techniques has gained increasing popularity among practitioners, largely due to the significant barrier that time constraints pose to training program adherence (45). This article will provide a comprehensive overview of the theories and mechanisms behind these advanced training strategies for enhancing muscle growth and provide practical guidelines for bodybuilders and fitness enthusiasts seeking to incorporate these techniques effectively into their training programs.

SUPERSETS

Supersets are an advanced resistance training method where 2 or more exercises are performed successively with relatively short or no rest between them. Although supersets are typically classified as a single technique, they can be further divided into specific types. Superset subdivisions include agonist/antagonist, upper body/lower body, compound agonist to isolation or compound agonist, isolation agonist to compound agonist (e.g., pre-exhaustion supersets, which will be discussed later), tri-sets, and giant sets. Table 1 provides a brief explanation of the various superset techniques, all of which share the

KEY WORDS:
supersets; drop sets; lengthened partials; rest-pause training; forced repetitions; blood flow restriction

common goal of enhancing training density by increasing workout volume within a shorter time frame compared with traditional strength training methods.

Recently, Burke et al. (13) investigated the effects of agonist/antagonist supersets as compared with a traditional resistance training program on muscle strength, hypertrophy, body composition, and endurance in resistance-trained men and women. The training protocol included the lat pull-down, bench press, seated leg curl, leg extension, dumbbell biceps curl, and cable triceps pushdown. All exercises were performed twice per week in the same session, and participants performed 4 sets of each exercise to momentary muscular failure (MMF: the point during a set when no additional repetitions can be completed with proper form despite maximal effort) with 8–12 repetition maximum (RM) loads. Participants in the traditional group completed all sets for 1 exercise before performing a different exercise with 2 minutes of rest between sets, whereas participants in the superset group performed exercises in an agonist/antagonist muscle group fashion (i.e., lat pull-down followed immediately by bench press) with 2 minutes of rest between each superset until a total of 4 sets were performed per superset. After 8 weeks, the authors reported similar increases in muscle thickness across all assessed muscle groups and similar changes in strength, power, local muscular endurance, and body composition outcomes. Despite comparable changes in muscular adaptations, the superset group completed the training sessions in 36% less time as compared with the traditional group. Hence, the findings of this study support the use of agonist/antagonist supersets as a time-efficient strategy to obtain favorable, but not superior, gains in muscle hypertrophy and strength compared with traditional training methods.

Iversen et al. (46) examined the effectiveness of supersets during full-body resistance training by assigning untrained participants to either a superset or traditional set group. Both groups performed 10 weeks of heavy resistance training twice per week consisting of the leg press, bench press, lat pull-down, and seated rows, but the superset group performed the exercises in both agonist to antagonist fashion (i.e., bench press to seated row) and lower-body to upper-body fashion (i.e., leg press to lat pull-down). Both groups experienced comparable improvements in body composition parameters and gained similar strength in the leg press and bench press. However, the traditional group showed a significantly greater improvement in the lat pull-down (+5.2 kg) compared with the superset group, along with a trend toward a greater improvement in seated row strength (+4.8 kg, $p = 0.073$). Although both groups similarly improved strength and body composition, the authors concluded that superset training may have limited maximal strength gains in exercises performed second within a superset sequence (e.g., rows and lat pull-downs) in the untrained cohort.

Table 1
Incorporating supersets into a bodybuilding program

Superset technique	Description	Example
Agonist → antagonist supersets	Perform an exercise for a muscle group. Rest 30–60 s and then perform an exercise for the antagonist muscle(s). Rest 30–60 s and repeat 2–5 times	Romanian deadlift Rest 60 s Seated leg extensions Rest 60 s (repeat 2–5 times)
Upper → lower body supersets	Perform an exercise for the upper body. Rest 30–60 s and then perform a lower body exercise. Rest 30–60 s and repeat 2–5 times	Dumbbell chest press Rest 60 s Reverse barbell lunges Rest 60 s (repeat 2–5 times)
Compound agonist → isolation or compound agonist	Perform an exercise for a muscle group. Rest 5–60 s and then perform an isolation or compound exercise for the same muscle(s). Rest 30–120 s and repeat 2–5 times	Dumbbell shoulder press Rest 30 s Dumbbell lateral raise or dumbbell upright row Rest 60 s (repeat 2–5 times)
Tri-sets and giant sets	Perform a series of 3 (tri-sets) or ≥ 4 (giant sets) isolation or compound exercises for a muscle group with 5–20 s rest between sets. Rest 30–120 s and repeat 2–4 times	Barbell shoulder press Rest 10 s Dumbbell lateral raises Rest 10 s Dumbbell front raise Rest 10 s Dumbbell upright rows (rest 90 s and repeat 3 times)

In addition, Demirtas et al. (22) investigated the effects of modified German volume training, compound agonist to isolation or compound agonist (depending on the muscle group) superset training, and giant set training on upper-body muscle hypertrophy and strength in recreationally trained men (i.e., participants with at least 1 month of resistance training adaptation). Although this study did not include a direct comparison to a traditional training protocol, all 3 groups increased the cross-sectional area (CSA) and thickness of the pectoralis major, deltoid, and latissimus dorsi muscles. Notable strength gains were also observed in the bench press, barbell row, and shoulder press exercises after 6 weeks of training. However, the researchers found no significant differences in muscle CSA, muscle thickness, or strength gains between the groups.

In summary, most evidence supports superset training as a time-effective strategy for increasing muscle hypertrophy and strength, yet this training technique is not superior to traditional training (106). Superset training does not negatively impact muscle activation, perceived recovery, or muscular adaptations when compared with traditional resistance training (106). Superset training has also been shown to maintain workout quality, as the training volume achieved during a workout can be comparable to that of traditional training (5,79). Nevertheless, supersets generally induce higher internal loads and energy cost along with an increased perceived exertion (106). Ultimately, a major advantage of superset training lies in its time-efficiency, offering a practical and effective option for individuals with limited time to train–without compromising training outcomes.

DROP SETS

Drop sets are an advanced resistance training technique, which is performed by taking sets to MMF at a given load followed by 1 to 3 "drops" where the load is reduced by approximately 20–25% and the exercise is immediately performed again to MMF (Table 2). Drop sets may be particularly effective for enhancing hypertrophy, as they involve multiple sets taken to failure within a short time frame. Several studies have investigated the effects of drop sets in comparison to traditional sets on measures of muscle hypertrophy (7,24,30–32,50,67,97). A recent systematic review and meta-analysis reported no significant difference in muscle growth between drop set and traditional training groups. However, drop sets were found to be a time-efficient strategy for inducing hypertrophy because a volume-equated drop set protocol takes half to one-third of the time compared with traditional training (90). For example, Angleri et al. (7) had resistance-trained men perform unilateral leg press and leg extensions using either a traditional protocol (3–5 sets of 6–12 repetitions at 75% 1RM with 2-minute rest between sets) or a volume-equated drop set protocol (3–5 sets of ~50–75% 1RM to MMF) twice weekly for 12 weeks. Participants performed up to 2 "drops" (reduction of 20% of load) after initially reaching MMF on each set. If the predetermined target volume for each exercise was reached before the end of the second drop (e.g., the first drop of the second set), the exercise was terminated to ensure the equalization of volume with the traditional protocol. Participants after the drop set protocol did not gain more strength or experience greater increases in vastus lateralis CSA compared with those in the traditional protocol. Enes et al. (24) had resistance-trained men perform the back squat, leg press, leg curl, leg extension, and stiff legged deadlift for 8 weeks using either a traditional (4 sets of 12 repetitions) or volume-equated drop set protocol (3 sets of 10 repetitions plus 1 drop set of 6 repetitions). Similarly, the drop set protocol failed to produce superior strength gains or greater increases in muscle thickness of the proximal, middle, and distal portions of the lateral thigh. Interestingly, Varovic et al. (97) had recreationally trained men follow either a traditional or drop set unilateral leg extension protocol without equating volume and reported that the drop set group experienced greater increases in rectus femoris muscle thickness at 30 and 50% muscle length (Δ%: 17 versus 3.7%, 8.3 versus 3.6%; Δ: 0.45 versus 0.10 cm, 0.23 versus 0.10 cm, respectively).

Current evidence suggests that drop sets do not outperform traditional sets for building muscle, yet they offer a time-efficient strategy. Substituting traditional sets with volume-equated drop sets significantly reduces workout duration without compromising muscle growth. In addition, incorporating drop sets into a training session effectively increases training volume without substantially extending workout

Table 2
Incorporating drop sets into a bodybuilding program

Strategy	Description	Example
Drop set	Perform set to MMF at a given load followed by 1 to 3 "drops" where the load is reduced by 20–25% and the exercise is immediately performed again to MMF	Perform an initial 10RM on cable curls using 120 lbs. (54.4 kg). Reduce weight to 95 lbs. (43.2 kg) and immediately perform a set to MMF. Reduce weight to 75 lbs. (34.1 kg) and immediately perform another set to MMF.

10RM = 10 repetitions maximum; MMF = momentary muscular failure (the point during a set when no additional repetitions can be completed with proper form despite maximal effort).

duration. Drop sets may be particularly effective with machine-based exercises due to the enhanced stability, lower injury risk, and reduced need for a spotter. Researchers still need to examine how multiple "drops" in the final set of an exercise (as traditionally used by bodybuilders) affect training outcomes. Furthermore, it is important to investigate whether drop sets lead to increased muscle hypertrophy under the following conditions: (a) when used in a real-world training setting without controlling for total volume load, (b) when compared with a traditional training method that matches the volume of drop sets, and (c) by taking into account how overall training volume affects hypertrophy outcomes, rather than solely comparing different set structures.

LENGTHENED PARTIALS

Range of motion (ROM) can operationally be defined as the degree of movement occurring at a given joint during an exercise. Many studies have reported that training with full ROM or at longer muscle lengths utilizing partial ROM (e.g., biceps curl only in the initial portion of the concentric muscle action) produce greater muscle hypertrophy compared with training at shorter muscle lengths (e.g., biceps curl only in final portion of the concentric muscle action) (10,52,55,56,59,60,71,72,94,105). Therefore, it has been proposed that exercises should be performed with a full ROM if the goal is to achieve maximal muscle hypertrophy (68). Yet, several studies have compared partial ROM training at longer muscle lengths to full ROM on muscle hypertrophy and reported either greater (48,71) or comparable outcomes (40,101,103). For example, Kassiano et al. (48) reported greater muscle hypertrophy of the medial gastrocnemius after performing calf raises for 8 weeks with a partial ROM at longer muscle lengths (Δ%: 15.2, Δ: 0.23 cm) compared with full ROM (Δ%: 6.7, Δ: 0.10 cm) and partial ROM at shorter muscle lengths (Δ%: 3.4, Δ: 0.06 cm) in young untrained women. The researchers also observed a greater increase in lateral gastrocnemius thickness in the group using partial ROM at longer lengths (Δ%: 14.9, Δ: 0.24 cm) compared with the full ROM group (Δ%: 7.3, Δ: 0.12 cm), but the difference did not reach statistical significance. A notable finding was that volume load progression was greater for both partial ROM groups compared with the full ROM group (48). A recent study by Gschneidner et al. (40) compared the effects of training with a full ROM and partial ROM at longer muscle lengths on muscle CSA of the upper arm and lower thigh. After 12 weeks, authors reported that both conditions will likely result in small and practically equivalent effects on muscle hypertrophy (40). A limitation of the study is that the researchers estimated CSA using circumference and skinfold measurements rather than a direct measurement such as ultrasound.

It is also well documented that muscle growth can occur in a nonuniform or region-specific manner (63). Several studies examining the effects of ROM on muscle hypertrophy have measured changes (e.g., muscle thickness or CSA) in only 1 region of the muscle, typically the mid-belly (39,48,56,65,73,101). However, previous studies suggest that training at longer muscle lengths may promote greater growth of the distal regions of the muscle (71,72,85). For example, Sato et al. (85) compared hypertrophy of elbow flexors after training at longer versus shorter muscle lengths, finding greater increases at distal regions (12.8%) compared with mid-belly (7.1%) and proximal regions (5.4%) after longer muscle length training. Limiting measurements to 1 region may prevent researchers from capturing the full effectiveness of a training protocol. For bodybuilders focused on maximizing muscular development and symmetry, a region-specific approach could offer meaningful insights (26). Although some individual studies have reported greater growth in the distal regions, a recent preprinted meta-analysis (96) found that both shorter and longer muscle length training resulted in similar regional hypertrophy across the proximal, mid-belly, and distal regions. The findings suggest no clear advantage of training at longer muscle lengths for regional hypertrophy, although the small difference in muscle lengths employed between conditions limits the strength of this conclusion.

A recent meta-analysis (102) reported trivial differences in muscle hypertrophy when comparing partial ROM and full ROM. Subanalyses of studies comparing partial ROM at longer muscle lengths versus full ROM suggest that partial ROM at longer muscle lengths results in a small benefit for muscle hypertrophy with a higher degree of uncertainty (standardized mean difference [SMD]: −0.28 [−0.81 to 0.16]). In contrast, this effect is not observed for partial ROM at shorter muscle lengths (SMD: 0.08 [−0.24 to 0.42]). However, caution is warranted, as the effect estimates were imprecise, indicated by wide 95% confidence intervals that include the null estimate, and the subanalyses included only 8 studies from both trained and untrained cohorts, highlighting the need for further research on the topic. Interestingly, Oranchuk et al. (66) indicated in a systematic review that isometric training (where no movement occurs through a ROM) at longer muscle lengths results in greater hypertrophy than at shorter lengths. The current evidence regarding the potential superiority of partial ROM at longer muscle lengths over full ROM remains equivocal. Yet, none of the studies reported negative impacts on hypertrophy from partial ROM training at longer muscle lengths, suggesting that this approach can be safely integrated into a training routine and may offer potential benefits.

Based on the current evidence, there are several ways in which lengthened partials can be implemented in resistance training programs (Table 3). Lengthened partials may be performed on each set of a given exercise; however, practitioners should

Table 3 Incorporating lengthened partials into a bodybuilding program		
Lengthened partials technique	**Description**	**Example**
Lengthened partials	Perform partial ROM repetitions in the lengthened position for all sets or the final 1–2 sets for specific, well-suited exercises	Perform lengthened partials for variations of rowing and pulling type exercises, biceps curls, knee extensions, or calf raises
Lengthened supersets (extended set)	Perform additional partial ROM repetitions in the lengthened position after reaching MMF	Perform set of full ROM standing calf raises to MMF and immediately continue performing partial repetitions in the lengthened position to MMF

MMF = momentary muscular failure (the point during a set when no additional repetitions can be completed with proper form despite maximal effort); ROM = range of motion.

carefully consider which exercises are suitable for lengthened partials. Research on lengthened partials has thus far been limited to a few muscle groups, such as the elbow flexors, elbow extensors, quadriceps femoris, and calves. In principle, trainees should aim to perform multiple exercises, which sufficiently lengthen or challenge the target muscle at longer muscle lengths; however, the question remains whether reaching maximal muscle lengths would confer even greater adaptations or whether there may be a point of diminishing returns. If an exercise has a resistance profile that is more challenging in the shortened portion, such exercises (e.g., knee extensions, calf raises, biceps curls) might be a viable option for lengthened partials. For these exercises, the extended time under tension during the lengthened phase of the movement may amplify mechanical tension, potentially enhancing the hypertrophic response. Conversely, performing lengthened partials on multijoint (MJ) exercises (e.g., squats, deadlifts) might not confer additional benefits for muscle hypertrophy, as these portions of the ROM are already sufficiently challenging at longer muscle lengths and may not result in increased mechanical tension. Nevertheless, researchers need to conduct more studies on lengthened partials across different exercises.

Lengthened partials can also be performed immediately after MMF as an "intensity technique" and a means to extend the set, subsequently increasing training volume and mechanical tension. Adding lengthened partials after reaching MMF may further enhance mechanical tension by prolonging exposure to higher training intensities and volumes, ultimately promoting a greater hypertrophic response. Indeed, Larsen et al. (53) found that performing lengthened partials after reaching MMF resulted in greater muscle hypertrophy (Δ%: 9.6, Δ: 0.19 cm) of the gastrocnemius compared with ending the set after reaching MMF (Δ%: 6.7, Δ: 0.14 cm) using a within-participant study design and untrained men. It is worth noting that the differences between groups (~0.04 cm) were very close to the measurement error reported by the authors (0.037–0.039 cm). Although this approach offers a time-efficient strategy to enhance training volume and intensity, it remains uncertain whether the benefits of lengthened partials after reaching MMF are driven by a specific mechanism or merely by an increase in training volume. Finally, one could consider alternating between partial ROM at different muscle lengths and full ROM across training sessions. Alternating ROM styles could produce distinct regional hypertrophy adaptations, as recently demonstrated by Pedrosa et al. (71) who examined the impact of training through various ranges of motion on muscle hypertrophy along the length of the quadriceps femoris in untrained women.

Although lengthened partials represent a promising training strategy for promoting muscle hypertrophy and bodybuilders can safely incorporate them into their training routines, the current body of research is mostly based on untrained cohorts and specific settings. As a result, the evidence remains inconclusive regarding their advantages compared with full ROM training. Notably, no study to date has shown full ROM to outperform partial ROM in this context. Further research is needed to investigate the effects of lengthened partials on other muscle groups and to establish best practices for incorporating them into an otherwise well-rounded bodybuilding routine. From a methodological standpoint, researchers must account for total training volume performed, time spent in the lengthened phase when comparing different ROMs, and finally, the novelty of the stimulus, particularly in individuals accustomed to exercises biased toward either the shortened or lengthened position. These factors are essential for advancing our understanding of this technique and developing practical recommendations for physique athletes.

PRE-EXHAUSTION TRAINING

General resistance training guidelines typically suggest performing MJ exercises before single-joint (SJ) exercises within a session. However, pre-exhaustion training, which is a specific type of superset, reverses this order by having individuals first perform a SJ exercise that isolates the agonist muscle, followed immediately by a MJ exercise targeting the same muscle group (e.g., pec deck before bench press). This approach aims to increase mechanical and metabolic stress, training volume/density, and muscle activation during the MJ exercise, thereby enhancing muscle growth of the target muscles (76). Alternatively, in "reverse pre-exhaustion," the SJ exercise targets a synergist of the MJ movement, potentially reducing its contribution to the MJ exercise and increasing stress on the agonist muscle group (e.g., triceps pushdown before bench press) (76). Table 4 presents examples of pre-exhaustion training strategies. Research has examined the effects of pre-exhaustion strategies on acute performance outcomes, neuromuscular activity, and chronic muscular adaptations (92).

Some studies (28,61,78) indicate that a tri-set pre-exhaustion approach–performing SJ exercises before MJ exercises–may facilitate a greater volume load compared with the traditional order. For example, Faria et al. (28) demonstrated that performing 3 sets of lower-body exercises to MMF at 75% 1RM resulted in a greater number of repetitions and higher training volume when exercises were performed in the order of knee extension, leg press, and squat, compared with the reverse order (squat, leg press, knee extension). Similar findings were observed for upper-body exercises, with a sequence of pec deck, incline bench press, and bench press yielding superior training volume compared with the inverse order (78). Alternatively, other studies indicate that the pre-exhaustion method of combining a SJ exercise immediately followed by a MJ exercise decreases the volume achieved in the MJ exercise (8,12,38,89,98). For example, performing dumbbell flys to MMF immediately before a set of bench press significantly reduced bench press repetitions (12), and performing knee extensions to MMF immediately before a set of leg press significantly reduced leg press repetitions (8), using a 10RM load. Likewise, Gentil et al. (38) had resistance-trained men perform 1 set of pec deck to MMF followed by 1 set of chest press to MMF, or vice versa, using a 10RM load. When the pec-deck was performed first, participants completed fewer repetitions during the chest press, and when the chest press was performed first, participants completed fewer repetitions during the pec deck. However, there was no difference in total work or total repetitions between conditions (38).

Most studies show that pre-exhaustion with an agonist SJ exercise does not increase electromyography (EMG) activity of the target muscle during MJ exercises (12,34,38,89). One study even found that performing knee extensions to MMF immediately before leg press significantly reduced rectus femoris and vastus lateralis activation during the leg press (8). However, performing a SJ exercise for the pectorals before completing a MJ chest exercise increased triceps activation suggesting greater triceps contribution when the pectorals were pre-exhausted, but no increased activation of the pectorals themselves (12,38). Nevertheless, acute studies cannot assess the long-term effects on muscle hypertrophy, making longitudinal studies essential.

First, several longitudinal studies have examined the impact of resistance exercise order on strength and hypertrophy gains, mostly in untrained individuals. A recent meta-analysis showed that trainees can achieve similar muscle growth regardless of exercise order, but they gain more strength in exercises performed at the beginning of the resistance training session (64). Next, few studies have assessed long-term muscular adaptations with the use of a pre-exhaustion method as compared with a traditional protocol. Aguiar et al. (1) reported significant improvements in CSA of the quadriceps muscle with pre-exhaustion compared with traditional training in untrained men. Both groups followed the same knee extension protocol (2 days per week; 3 sets of 8–12

Table 4
Incorporating pre-exhaustion training into a bodybuilding program

Pre-exhaustion training technique	Description	Example
SJ agonist → MJ exercise for same muscle group	Perform SJ exercise for agonist muscle immediately before MJ exercise for same muscle group	Pec deck (20–50% 1RM) to MMF followed immediately by bench press (~75% 1RM)
SJ synergist → MJ exercise (i.e., reverse pre-exhaustion)	Perform SJ exercise for synergist muscle immediately before MJ exercise	Triceps pushdown (20–50% 1RM) to MMF followed immediately by bench press (~75% 1RM)
SJ-to-MJ multiple-set sequence	Perform all sets of SJ exercise before sets of MJ exercise	3 sets of pec deck before 3 sets of bench press

1RM = 1 repetition maximum; MJ = multijoint; MMF = momentary muscular failure (the point during a set when no additional repetitions can be completed with proper form despite maximal effort); RM = repetition maximum; SJ = single-joint.

repetitions at 75% 1RM) for 8 weeks, with the pre-exhaustion group adding 1 additional set of knee extensions using 20% 1RM to MMF 30 seconds before each session. Others have failed to show benefit of pre-exhaustion techniques on measures of muscle hypertrophy (33,50,93). For example, Trindade et al. (93) found no difference in quadriceps muscle thickness when an additional set of leg extensions (20% 1RM to MMF) was performed immediately before 3 sets of leg press to MMF at 75% 1RM (twice weekly for 9 weeks) in untrained men. Most recently, Keskin et al. (50) found no difference in quadriceps muscle thickness after a 6-week (2 days per week) traditional versus pre-exhaustion training program in recreationally trained men. Both groups performed 8 sets of leg press (70% 1RM; 8–12 repetitions) with 3-minute rest intervals to match training volume. The pre-exhaustion group replaced the first and fourth sets with leg extensions to MMF at 30% 1RM.

Much of the research did not apply the pre-exhaustion method as intended, using lighter-load SJ exercise (20–50% 1RM) followed immediately by a MJ exercise with a heavier load (~75% 1RM). Therefore, it is important to consider not only the findings from individual studies but also the specific pre-exhaustion routine that was implemented (e.g., exercises, load, and transition interval) when drawing conclusions from the literature. Although performing a lighter-load SJ exercise immediately prior tends to decrease the number of repetitions completed during the MJ exercise, an additional SJ exhaustive set before MJ movements will likely increase total training volume and metabolic stress on the target muscle as compared with traditional training without the pre-exhaustive set. If the goal is to reach a certain number of MJ repetitions when using pre-exhaustion training, the load in the MJ exercise may need to be reduced (a strategy that also reduces forces applied to the joints) (92). Ultimately, current evidence on pre-exhaustion training remains limited and inconclusive regarding its effects on muscle hypertrophy. Most of the available data is acute and there remains a lack of longitudinal studies implementing the pre-exhaustion method.

REST-PAUSE TRAINING

Rest-pause training typically involves lifting a fixed load (40–80% 1RM) with an initial set to MMF, followed by subsequent sets to MMF using short intraset rest intervals (typically 10–20 seconds) (24,35,58,74). By incorporating brief intraset rest periods, this method may help individuals complete a target number of repetitions at a given load (Table 5). Rest-pause training may be a time-saving and effective strategy for promoting skeletal muscle hypertrophy, as it allows for increased training volume within a given set while sustaining the same load (74).

The rest-pause method may enhance muscle activation while producing similar levels of acute neuromuscular fatigue as traditional protocols. Marshall et al. (58) examined the acute neuromuscular and fatigue responses in resistance-trained males to 3 different training approaches (1): a rest-pause method (initial set to failure at 80% 1RM with subsequent sets performed with 20 seconds interset rest intervals for a total of 20 squat repetitions) (2), a short rest method (5 sets of 4 squats at 80% 1RM with 20 seconds interset rest intervals), and (3) a traditional method (5 sets of 4 squats at 80% 1RM with 3-minute interset rest intervals). As expected, all protocols similarly decreased maximal force and rate of force development immediately after completion. However, the rest-pause method demonstrated greater muscle activation compared with the other protocols. In addition, the rest-pause method required substantially less time to complete (103 seconds) compared with the short rest method (140 seconds) and the traditional method (780 seconds) (58).

When total training volume is matched, rest-pause training has been evaluated against traditional training

Table 5
Incorporating rest-pause into a bodybuilding program

Rest-pause technique	Description	Example
Rest-pause (preplanned repetitions)	Perform sets to MMF at a given load with short rest intervals (15–20 s) between sets until reaching a preplanned total repetition goal	Rather than performing 3 sets of 8RM (24 repetitions) on machine shoulder press, perform the first set to MMF, then rest 15–20 s and complete an additional set to MMF. Repeat until 24 repetitions are completed
Rest-pause (extended set)	Perform initial set to MMF at a given load. Take a short rest interval (15–20 s) and repeat another set to MMF with the same load. Rest another 15–20 s and repeat 2–3 times	Perform a set to MMF on the chest press machine using ~70–80% 1RM Rest 15–20 s Perform another set to MMF at the same load Rest 15–20 s (repeat 2–3 times)

1RM = 1 repetition maximum; MMF = momentary muscular failure (the point during a set when no additional repetitions can be completed with proper form despite maximal effort).

methods for its effects on hypertrophy and strength (24,47,74). Prestes et al. (74) compared the longitudinal effects of a 6-week full-body training program using either rest-pause (1 initial set to MMF at 80% of 1RM with subsequent sets performed with a 20-second interset rest interval until completing a total of 18 repetitions) or traditional training (3 sets of 6 repetitions with 80% of 1RM and a 2-minute interset rest interval) in trained subjects. Before and after the training program, the researchers evaluated body composition (via skinfold measurements), muscle strength (1RM), and endurance (RM sets) for the bench press, leg press, and free-weight standing biceps curl. Muscle thickness of the arm, thigh, and chest were assessed using ultrasound. The rest-pause group achieved significantly greater localized muscular endurance for the leg press only, with no difference in body composition or strength. No significant differences were observed in the percentage change in arm and chest muscle thickness. However, thigh muscle thickness increased significantly more with rest-pause training (11 ± 14%) compared with traditional training (1 ± 7%). Notably, the authors failed to report specific measurement sites or measurement error and presented results only as percentage changes without providing raw values. Next, Enes et al. (24) compared the effects of lower body rest-pause training and traditional training over 8 weeks, finding that although rest-pause training led to superior strength adaptations, it did not result in superior hypertrophic adaptations when total training volume was equalized. In a recent study by Karimifard et al. (47), 12 weeks of rest-pause training led to greater increases in both strength and hypertrophy compared with traditional training in untrained men. However, thigh and arm CSA were estimated using circumference measurements, the reported effect sizes were unusually large, and measurement error was not provided.

Bodybuilders can use rest-pause training as a time-efficient way to enhance skeletal muscle hypertrophy. The combination of failure-driven sets, short interset rest periods, and increased training volume within a compressed timeframe may be beneficial for promoting muscle growth. Bodybuilders may find rest-pause sets to be well-suited for machine-based training because of the greater stability, lower injury risk, and less reliance on a spotter. However, although some studies have favored rest-pause training, they contain methodological issues that must be accounted for, such as estimation errors in muscle size, lack of volume-load equalization, and absence of blinding procedures. Notably, there is a lack of research that does not control for total training volume, a context where rest-pause could excel by facilitating more quality volume within a single session compared with traditional protocols. Although this has practical importance, accounting for total training volume is essential to advancing our understanding of this technique. Overall, more high-quality research, particularly in competitive bodybuilders, is needed to draw more precise conclusions regarding its long-term effectiveness and applicability across diverse populations and training contexts. Furthermore, no studies have examined the strategic integration of rest-pause methods within the framework of a comprehensive training program, leaving gaps in understanding its long-term potential for optimizing hypertrophy outcomes.

FORCED REPETITIONS

Forced repetitions is an advanced resistance training strategy that involves a training partner providing assistance to complete additional repetitions immediately after reaching MMF (Table 6) (41). Despite the widespread popularity among bodybuilders, researchers have not thoroughly examined their long-term effects on muscle growth. Some studies have examined acute responses with the use of forced repetitions as compared with sets only taken to MMF (2–4,37). In a series of studies, Ahtiainen et al. (2–4) compared a traditional training protocol, in which sets were performed to MMF using a 12RM load, with a forced repetition protocol in strength-trained or physically active/recreationally trained men. In the latter, a heavier load was used so that the subject could only complete approximately 8 repetitions unassisted, necessitating the use of forced repetitions to reach the 12 repetitions per set. Forced repetition protocols increased time under tension and led to greater postexercise declines in isometric strength, indicating a higher level of neuromuscular fatigue (2–4). Blood lactate (2–4), creatine kinase (4), muscle thickness (4), and muscle soreness ratings (4) were similar between traditional and forced repetition protocol. The forced repetition protocol led to a greater acute increase in cortisol (4) and growth hormone (4), yet both protocols similarly increased testosterone and free testosterone levels (3,4). In addition, Wallace et al. (100) reported that 2 bouts of forced repetitions performed within 48 hours resulted in similar volume load, muscle activation, perceived exertion, and metabolic stress compared with traditional training in resistance-trained individuals.

Only 1 study has included forced repetitions in a longitudinal resistance training intervention (23). In this study, 12 male basketball players and 10 male volleyball players participated in a 6-week training program, performing the bench press 3 times per week. Participants followed a set-and-repetition scheme of either 4 × 6, 8 × 3, or 12 × 3, each specifically designed to produce varying numbers of forced repetitions per training session. Despite differences in the number of forced repetitions performed during training, all groups showed similar increases in chest circumference (0.5 cm) and estimated muscle mass (0.6 kg), with no significant differences between groups. Although statistically significant, these changes were likely not practically meaningful because both fell below the smallest worthwhile change thresholds (0.9 cm for chest circumference and 1.1 kg for estimated muscle mass). Beyond the lack of a positive

Table 6
Incorporating forced repetitions into a bodybuilding program

Strategy	Description	Example
Forced repetitions	Perform set to MMF then immediately have a training partner provide assistance to complete 3–5 additional repetitions	Perform a final set of chest-supported rows at ~60–80% 1RM to MMF. Training partner then immediately provides assistance to complete 3–5 additional repetitions

1RM = 1 repetition maximum; MMF = momentary muscular failure (the point during a set when no additional repetitions can be completed with proper form despite maximal effort).

control group performing traditional resistance training, the study employed a rudimentary approach to assessing muscle size changes, relying on anthropometric estimates rather than direct imaging techniques. This low sensitivity approach introduces measurement noise, making it difficult to detect subtle but meaningful changes in muscle hypertrophy. Therefore, more studies are needed to accurately assess changes in muscle hypertrophy in trained cohorts performing forced repetitions compared with traditional resistance training.

Current research does not provide enough evidence to determine whether incorporating forced repetitions enhances muscle hypertrophy compared with traditional resistance training or other advanced training strategies. The absence of well-controlled longitudinal studies limits our ability to offer practical recommendations at this time. Further research is needed to evaluate its effectiveness and adherence, particularly given its low practicality compared with other techniques, as forced repetitions require a spotter. Until more robust data are available, practitioners aiming for muscle hypertrophy should consider whether the logistical challenges of forced repetitions, such as the requirement of a spotter, outweigh the benefits of other better-supported training methods. Forced repetitions may also elevate injury risk if technique and form are compromised due to significant fatigue.

PYRAMIDAL TRAINING

Pyramidal training is a method that systematically adjusts both repetitions and load across successive sets of a given exercise in a resistance training session. Because intensity is modified on a set-to-set basis, a trainee can adjust the repetition range used in the way that best aligns with their training goals. Pyramidal training can be structured in 2 primary ways: ascending (or crescent) or descending (or decrescent) pyramids (Table 7). In the ascending pyramid, load increases while the number of repetitions decreases across sets. Conversely, in the descending pyramid, load decreases while the number of repetitions increases with each subsequent set (15). In addition, one can perform a "full pyramid," which combines both the ascending and descending approaches into an extended series of sets. The progressive adjustments in load and repetition schemes alter both force and metabolic demands across working sets, and many consider this approach optimal for simultaneously building strength and hypertrophy. Although many bodybuilders use pyramidal training, researchers have

Table 7
Incorporating pyramidal training into a bodybuilding program

Pyramidal training technique	Description	Example
Ascending pyramid protocol	Load increases while the number of repetitions decreases as sets progress	Set 1: 12 repetitions at 65% 1RM Set 2: 10 repetitions at 70% 1RM Set 3: 8 repetitions at 75% 1RM
Descending pyramid protocol	Load decreases while the number of repetitions increases as sets progress	Set 1: 8 repetitions at 75% 1RM Set 2: 10 repetitions at 70% 1RM Set 3: 12 repetitions at 65% 1RM
Full pyramid protocol	Combining the ascending and descending approach into an extended series of sets	Set 1: 12 repetitions at 65% 1RM Set 2: 10 repetitions at 70% 1RM Set 3: 8 repetitions at 75% 1RM Set 4: 10 repetitions at 70% 1RM Set 5: 12 repetitions at 65% 1RM

1RM = 1 repetition maximum.

not confirmed its superiority over traditional training.

Several studies have investigated how pyramidal training affects acute physiological and subjective responses compared with traditional training (6,17,18,25,57,70). Angleri et al. (6) observed comparable muscle activation (measured through surface EMG amplitudes) and microvascular oxygenation (assessed via near-infrared spectroscopy) during a leg extension ascending pyramidal protocol (10, 8, and 6 repetitions at 75, 80, and 85% 1RM) and a traditional protocol (3 sets of 10 repetitions at 75% 1RM) in recreationally trained young men. In untrained young men, an upper body ascending pyramidal protocol (3RM sets with 67, 74, and 80% 1RM) produced similar acute responses in hormonal markers (testosterone, growth hormone, cortisol), metabolic indicators (glucose, lactate), and muscle damage markers (creatine kinase, C-reactive protein) compared with a volume-equated traditional protocol (3RM sets with 75% 1RM). Both protocols also elicited comparable levels of subjective muscle soreness and ratings of perceived exertion (RPE) (17,18). Similarly, in untrained elderly men, a lower-body ascending pyramidal protocol (12, 10, and 8 repetitions at 67, 75, and 80% 1RM) produced comparable hormonal and metabolic responses to a traditional protocol (3 sets of 10 repetitions at 75% 1RM) (70). In trained men, Enes et al. (25) found no differences in total training volume between a lower body descending pyramidal protocol (8, 10, and 12 repetitions at 75, 70, and 65% 1RM) and a traditional protocol (3 sets of 10 repetitions at 70% 1RM) consisting of barbell back squat, leg press, and leg extensions. However, the pyramidal protocol resulted in higher RPE and lower ratings of pleasure compared with the traditional protocol. Last, in well-trained lifters, Mang et al. (57) compared an ascending bench press pyramid (65, 70, 75, 80, and 85% 1RM), a descending pyramid (85, 80, 75, 70, and 65% 1RM), and a traditional protocol (5 sets with 75% 1RM). No differences were observed in total volume load, enjoyment, session-RPE, discomfort, or affect. However, peak velocity and set-RPE were significantly lower during the descending pyramid compared with the ascending pyramid and the traditional protocol.

Recent research has explored the long-term effects of pyramidal training strategies compared with traditional training protocols on muscular adaptations. Ribeiro et al. (77) compared an 8-week full-body training program performed 3 times per week after either traditional loading (3 sets of 8–12RM) or ascending pyramid loading (3 sets of 12, 10, and 8RM) in untrained older women using a crossover study design. No significant differences were observed in 1RM strength gains or skeletal muscle mass, as assessed by dual-energy x-ray absorptiometry. A follow-up study from the same lab (75) also found that both training systems effectively improved strength and muscle growth in resistance-trained older women, but that pyramid training did not outperform the traditional protocol. Similarly, in young trained men, ascending pyramid training (3–5 sets of 6–15 repetitions at 65–85% 1RM) did not result in greater improvements in strength and hypertrophy after a 12-week leg press and leg extension program when compared with a volume-equated traditional program (3–5 sets of 6–12 repetitions at 75% 1RM) using a within-subject design (7). Ultrasound assessments showed that vastus lateralis CSA increased significantly and similarly after both ascending pyramid training (7.5%) and traditional training (7.6%).

Current evidence suggests that pyramidal training elicits similar acute and chronic responses to traditional resistance training protocols. However, research on the pyramid technique has primarily focused on repetition ranges of approximately 5–15 repetitions within a narrow intensity range (65–85% 1RM). Researchers need to conduct more studies to explore how pyramidal training works across broader intensity ranges (e.g., 30–90% 1RM) and repetition schemes (e.g., 6–30 reps). Researchers should also conduct more studies in trained populations. Despite these gaps in the literature, pyramidal training remains one of the most versatile resistance training strategies, providing a practical strategy to introducing variety into a bodybuilding program.

Table 8
Incorporating interset stretching into a bodybuilding program

Interset stretching technique	Description	Example
Interset stretching (loaded stretch)	Perform loaded static stretching at the maximum pain-free range of motion for first 30 s of the rest interval (i.e., 30 s of stretching followed by 60 s of passive rest)	Performed a loaded stretch using the same load employed in the working sets immediately after sets of standing calf raises
Interset stretching (passive/active static stretching)	Perform active or passive static stretching at the maximum pain-free range of motion for first 30 s of the rest interval (i.e., 30 s of stretching followed by 60 s of passive rest)	Perform active or static stretching of the pectoralis muscles immediately after sets of bench press

INTERSET STRETCHING

One advanced training strategy that has recently garnered attention is interset stretching (ISS). ISS involves using various stretching methods during the rest period between working sets, typically applied to the primary mover of a given exercise. Because stretching is performed within the rest period (e.g., during a 90-second rest period, a 30-second stretch is followed by 60 seconds of passive rest), this technique does not extend the overall workout duration (Table 8). Despite its popularity, the literature on ISS has produced conflicting findings on muscle hypertrophy (14,27,95,99).

Regarding untrained populations, 2 studies have been conducted to date. Evangelista et al. (27) investigated the effects of ISS on the biceps, triceps, rectus femoris, and vastus lateralis, and the sum of these 4 muscle thickness sites, after an 8-week full-body RT program. Compared to the traditional group, the ISS group performed passive static stretching at the maximum pain-free ROM for 30 seconds during the 90-second rest interval (i.e., 30 seconds of stretching followed by 60 seconds of passive rest), whereas the traditional group rested for the full 90 seconds. Participants stretched the same muscle groups they trained during each set of the session. Post-hoc analyses showed that the vastus lateralis and the sum of muscle thickness improved more in the ISS group than in the traditional group (Δ%: 17 versus 7.3%, 10.5 versus 6.7%; Δ: 0.31 versus 0.24 cm, 0.94 versus 0.62 cm, respectively). In addition, using a within-group design, Van Every et al. (95) investigated the effects of ISS performed during calf raise exercises on muscle thickness in the lateral and medial gastrocnemius muscles, and the soleus, after an 8-week intervention. In the ISS group, participants performed a loaded stretch immediately after the completion of each set, using the same load employed in the working sets performed on seated and straight-leg calf raise exercises. The loaded stretch was held for 20 seconds, followed by passive rest for the remainder of the rest interval (20 seconds of stretching followed by 100 seconds of passive rest, totaling 120 seconds of rest between sets). Results indicated a potential hypertrophic benefit of ISS only for the soleus muscle compared with the traditional group (Δ%: 6.9 versus 3.5%; Δ: 0.12 versus 0.06 cm), with no clear effect observed in the gastrocnemius muscles.

In trained individuals, 2 studies have also been conducted up to this point (14,99). Wadhi et al. (99) examined the effects of ISS on muscle thickness of the pectoralis major at the belly and lateral portions during 8 weeks of training. The ISS group performed a loaded stretch during the rest period of working sets equivalent to ~15% of their working set load. As soon as subjects completed their working set on the bench press and/or incline bench press exercises, subjects moved directly to a cable machine and were instructed to hold a loaded cable-fly stretch (i.e., 30-second loaded stretch at 15% of previous set followed by 90-second passive rest). Findings indicated that the addition of loaded ISS did not augment muscle hypertrophy (∑MT–Δ%: 10.6 versus 10.4%; Δ: 0.59 versus 0.58 cm), nor reduce the total volume load in the trained population. Businari et al. (14) had recreationally trained individuals perform ISS for 8 weeks and examined biceps and triceps muscle thickness. However, the authors implemented a longer ISS duration (45 seconds), differing from previous studies that restricted ISS to 30 seconds or less. Despite the lower total training volume load in the ISS group, a small increase in muscle thickness was observed only in the distal portion of the triceps brachii after ISS (19.5 versus 9.5%; 0.30 versus 0.11 cm). It is noteworthy to mention that both ISS and the traditional group did not induce significant increases in muscle thickness.

To date, studies have primarily reported enhanced hypertrophic benefits of ISS in untrained participants, with underlying mechanisms yet to be fully elucidated (87). Furthermore, current evidence suggests that ISS-induced muscle mass accretion in trained cohorts remains uncertain or unlikely. However, key questions remain unanswered, such as whether the benefits of ISS are population-specific, particularly given the larger effects observed in certain demographics and muscle groups. In addition, does the effectiveness of ISS depend on muscle groups predominantly composed of type I fibers? Notably, no study to date has reported a reduction in hypertrophic adaptation, even when ISS led to a decrease in total volume load performed. Moreover, although beyond the scope of this review, ISS has not been shown to impair strength adaptations.

From a practical standpoint, ISS can be a viable tool in a strength coach's repertoire. However, nuances in the scientific literature are crucial for making informed programming decisions. Its application should be tailored to specific populations, muscle groups, and the individual preferences of the trainee to increase effectiveness.

BLOOD FLOW RESTRICTION

Recent evidence suggests that high external loads (>60–70% 1RM) are not a prerequisite for inducing muscle hypertrophy during resistance exercise, provided that training is performed with sufficient effort and near failure (54). Studies indicate that lighter loads (20–50% 1RM) can be effective alternatives to traditional high-load protocols (86). However, achieving comparable muscular adaptations with low loads typically requires strategies to enhance training stress, as the per-set repetition volume needed to approach failure with these loading schemes is very high (>50 repetitions per set using 20% 1RM) (81). BFR addresses this need by employing a compressive, typically single-chambered, tourniquet-like cuff applied to the most proximal portion of the limb. When inflated, the cuff restricts arterial inflow and venous return, thereby amplifying metabolic and physiologic stress at any given workload (11,51). This heightened stress is thought to allow individuals to reach MMF more rapidly compared with the same exercise without BFR, offering a practical means to optimize adaptations at lighter loads without use of high repetitions per set (51).

BFR training pressures are typically set as a percentage of limb occlusion pressure (LOP; 40–80%), which represents the minimum pressure required to occlude both venous and arterial blood flow (69). Evidence suggests a threshold for effective pressure application (≥40–50% LOP) (16,20), indicating that pressure may play a critical role in the effectiveness of the BFR training stimulus. Consequently, cuffs that allow for precise adjustments of applied pressure relative to LOP may be particularly beneficial, as they ensure the appropriate pressure is applied to achieve desired training outcomes in alignment with the literature. Multichambered bladder cuffs are available for consumer use; however, they are not capable of personalizing the applied pressure to the individual due to its design and follow distinct application guidelines (Figure 1) (80). Preliminary research indicates that during a typical 4-set BFR routine to MMF, multichambered cuffs inflated to manufacturer-recommended settings reduces the number of repetitions needed to achieve MMF relative to a no-BFR control, while still enabling significantly greater total exercise volume than single-chambered systems inflated to 60% LOP (81).

Current evidence suggests that multichambered cuffs may have practical utility, as they do not occlude arterial inflow and may be more suitable for nonclinicians due to their safety profile. Although their inferior effectiveness in accelerating fatigue accumulation and reducing exercise volume during low-load resistance training may limit their applicability for optimizing muscular hypertrophy, these cuffs could offer advantages for moderate loads (50–70% 1RM). The elastic nature of the cuff material may impose less vascular stress, making them a viable option for applications beyond traditional BFR resistance exercise while still enabling a reduction in exercise volume to achieve training intensities near failure. For further details, readers are referred to the following articles on the various types of BFR cuffs (83), methods of pressure prescription (9,19,69), and relevant safety screening guidelines and implementation strategies (62,69,82).

Figure 1. Multichambered versus single-chambered blood flow restriction (BFR) cuff designs. The top cuff represents a multichambered design, with vertical lines at the bottom of the cuff indicating individual air bladders. The bottom cuff illustrates a traditional single-chambered bladder design, consisting of a single air bladder extending the length of the cuff. When air is pumped into a single-chambered cuff, it applies uniform pressure around the limb, which can significantly reduce arterial inflow. This design allows users to determine arterial or limb occlusion pressure (LOP), facilitating a personalized approach to BFR exercise. In contrast, the multichambered cuff's design—with spaces between air bladders and the elastic nature of the material—prevents most users from accurately determining LOP. However, the pressure applied by multichambered cuffs can partially restrict venous return, enabling a form of BFR exercise without personalized pressure calibration. These cuffs are typically used at pressures ranging from 200 to 500 mm Hg, but this refers to the internal pressure within the cuff itself, not the actual pressure applied to the limb. Figure used with permission from (81).

Research has demonstrated that low-load BFR resistance exercise (20–50% 1RM) can elicit comparable muscular hypertrophic responses to high-load resistance exercise (>60% 1RM) in healthy adults (21) and trained athletes (36). Given that the primary goal of bodybuilders is muscle mass accretion, incorporating BFR into a bodybuilding program offers numerous advantages. Bodybuilders can use low-load BFR either in isolation or in conjunction with high-load resistance exercise during the same exercise session. Rolnick and Schoenfeld (84) outlined application guidelines for recreational and competitive bodybuilders that remain relevant and provide in-depth programming insights in addition to discussing the purported physiology of BFR. Since then, emerging evidence has highlighted BFR's ability to induce a potent postexercise hypoalgesia response, opening additional ways to leverage BFR in practice (43,44,91,104).

As bodybuilders frequently encounter musculoskeletal injuries (88), low-load BFR resistance exercise offers a dual benefit: facilitating muscle growth comparable to heavier loads while reducing irritation to injured body parts allowing training to continue. In addition, the associated local hypoalgesia response–which can also be systemic (43)–may enable bodybuilders to maintain high-intensity training for hypertrophy without relying on the heavy external loads typically used in standard training protocols. Beyond the aforementioned benefits, BFR presents a key practical advantage when heavy loads are contraindicated. Unlike traditional low-load resistance training, which necessitates a high number of repetitions to approach or reach MMF, BFR allows for muscle failure to be achieved with significantly fewer repetitions per set, thereby reducing overall workout time while still producing similar hypertrophic adaptations. Farup et al. (29) demonstrated that when low-load resistance training was performed to fatigue, BFR required roughly one-third of the total repetitions compared with traditional low-load training, yet both conditions resulted in equivalent muscle hypertrophy. Table 9 describes 2 evidence-based applications for a bodybuilder with knee pain that is load sensitive looking to improve quadriceps hypertrophy. All exercises are performed with 30 seconds of rest between sets and 2 minutes between exercises with the cuffs deflated.

CONCLUSION AND PRACTICAL RECOMMENDATIONS

The field of advanced resistance training strategies remains limited, and more well-controlled, long-term research is needed to determine the efficacy of these methods for promoting hypertrophy. Current studies in the field often consist of small sample sizes and short study durations (6–12 weeks). Lifters who regularly use advanced training strategies long term –especially those involving frequent training to failure–must account for the challenges of accumulated fatigue. In addition, many studies involve untrained or recreationally trained individuals, with none including competitive bodybuilders. There is also significant variability in study designs and training protocols. In addition, the observed effects in hypertrophy research, particularly in trained cohorts, are typically small, and measurement methodologies remain inconsistent and often lack transparency. Many studies fail to provide detailed imaging data, appropriate measurement error reporting, and clear methodological descriptions. These limitations hinder our ability to distinguish true differences from measurement noise, ultimately restricting the field's progress. Moreover, there is a lack of research exploring the strategic integration of these methods into an evidence-based bodybuilding program, which would offer greater ecological validity. Most studies investigate an all-or-nothing approach, comparing the use of only the advanced strategy versus a traditional protocol, often targeting a single muscle group. In addition, researchers often compare traditional training strategies with volume-equated advanced strategies to determine whether observed effects are due to factors related to the advanced strategy itself, as compared to merely increasing training volume. Although training volume is a key confounding variable when studying hypertrophy, this approach limits the ability to evaluate the true effectiveness of certain advanced strategies as they are originally intended to be used by bodybuilders. Because many advanced strategies are designed to help bodybuilders achieve higher training volumes and impose greater muscular overload in a time-efficient manner, further research is needed to enhance the real-world applicability of findings. Developing innovative experimental designs that enhance ecological validity ensuring these strategies are studied in a way that reflects real-world application while maintaining scientific rigor is essential (49).

In conclusion, many advanced strategies can be effective tools that offer novelty, improve time-efficiency, and increase training volume without significantly extending workout duration. Yet, their implementation requires careful consideration. In some cases, using these strategies continuously may lead to overtraining, psychological burnout, and overuse injuries due to the repeated demand on the neuromuscular system. Practitioners should integrate these strategies into a bodybuilding program based on the athlete's individual preferences, training experience, and recovery capacity. A strategic and periodized approach to advanced training strategies is essential for effectively inducing hypertrophic adaptations while minimizing potential drawbacks.

Table 9
Two evidence-based applications for blood flow restriction (BFR) training in a bodybuilder with knee pain

Application	Relevant BFR pressures	Programming suggestions
BFR in isolation (offloading injury)	50–80% LOP	1. BFR single leg press: 3–4 sets of 15 repetitions at 30% 1RM 2. BFR single leg extensions: 3–4 sets of 15 repetitions at 30% 1RM
BFR before heavy loading (hypoalgesic response to facilitate load tolerance)	80% LOP	1. BFR bilateral leg extensions: 1 set of 30 repetitions, 3 sets of 15/15/+ at 20% 1RM 2. Bilateral leg press: 3 sets of 10–12 repetitions at 60–75% 1RM

1RM = 1 repetition maximum; BFR = blood flow restriction; LOP = limb occlusion pressure; +: as many repetitions as possible.

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REFERENCES

1. Aguiar AF, Buzzachera CF, Pereira RM, et al. A single set of exhaustive exercise before resistance training improves muscular performance in young men. *Eur J Appl Physiol* 115: 1589–1599, 2015.
2. Ahtiainen JP, Häkkinen K. Strength athletes are capable to produce greater muscle activation and neural fatigue during high-intensity resistance exercise than nonathletes. *J Strength Cond Res* 23: 1129–1134, 2009.
3. Ahtiainen JP, Pakarinen A, Kraemer WJ, Häkkinen K. Acute hormonal responses to heavy resistance exercise in strength athletes versus nonathletes. *Can J Appl Physiol* 29: 527–543, 2004.
4. Ahtiainen JP, Pakarinen A, Kraemer WJ, Häkkinen K. Acute hormonal and neuromuscular responses and recovery to forced vs. maximum repetitions multiple resistance exercises. *Int J Sports Med* 24: 410–418, 2003.
5. Andersen V, Fimland MS, Iversen VM, et al. A comparison of affective responses between time efficient and traditional resistance training. *Front Psychol* 13: 912368, 2022.
6. Angleri V, de Oliveira R, Biazon TM, et al. Effects of drop-set and pyramidal resistance training systems on microvascular oxygenation: A near-infrared spectroscopy approach. *Int J Exerc Sci* 13: 1549–1562, 2020.
7. Angleri V, Ugrinowitsch C, Libardi CA. Crescent pyramid and drop-set systems do not promote greater strength gains, muscle hypertrophy, and changes on muscle architecture compared with traditional resistance training in well-trained men. *Eur J Appl Physiol* 117: 359–369, 2017.
8. Augustsson J, Thomeé R, Hörnstedt P, Lindblom J, Karlsson J, Grimby G. Effect of pre-exhaustion exercise on lower-extremity muscle activation during a leg press exercise. *J Strength Cond Res* 17: 411–416, 2003.
9. Bielitzki R, Behrendt T, Behrens M, Schega L. Current techniques used for practical blood flow restriction training: A systematic review. *J Strength Cond Res* 35: 2936–2951, 2021.
10. Bloomquist K, Langberg H, Karlsen S, Madsgaard S, Boesen M, Raastad T. Effect of range of motion in heavy load squatting on muscle and tendon adaptations. *Eur J Appl Physiol* 113: 2133–2142, 2013.
11. Bower M. Metabolic stress and blood flow restriction training as interventions for skeletal muscle atrophy following musculoskeletal injury. *Int J Strength Cond*. 2021;1:1–8.
12. Brennecke A, Guimarães TM, Leone R, et al. Neuromuscular activity during bench press exercise performed with and without the preexhaustion method. *J Strength Cond Res* 23: 1933–1940, 2009.
13. Burke R, Hermann T, Piñero A, et al. Less time, same gains: Comparison of superset vs traditional set training on muscular adaptations. *SportRx Preprint*. 2025.
14. Businari GB, Brigatto FA, de Camargo JBB, et al. Chronic effects of inter-set static stretching on morphofunctional outcomes in recreationally resistance-trained male and female. *Res Q Exerc Sport* 95: 10–23, 2024.
15. Cardozo DC, de Souza Destro D. Pyramidal resistance training: A brief review of acute responses and long-term adaptations. *J Bodywork Mov Therapies* 35: 21–27, 2023.
16. Cerqueira MS, Lira M, Barboza JAM, et al. Repetition failure occurs earlier during low-load resistance exercise with high but not low blood flow restriction pressures: A systematic review and meta-analysis. *J Strength Cond Res*. 2022. Published ahead of print.
17. Charro MA, Aoki MS, Coutts AJ, Araújo RC, Bacurau R. Hormonal, metabolic and perceptual responses to different resistance training systems. *J Sports Med Phys Fitness* 50: 229–234, 2010.
18. Charro MA, Aoki MS, Nosaka K, Foschini D, Figueira A Jr, Bacurau RF. Comparison between multiple sets and half-pyramid resistance exercise bouts for muscle damage profile. *Eur J Sport Sci* 12: 249–254, 2012.
19. Chulvi-Medrano I, Cortell-Tormo JM, Hernández-Sánchez S, Picón-Martinez M, Rolnick N. Blood flow restriction training in clinical rehabilitation: Occlusion pressure methods relative to the limb occlusion pressure. *J Sport Rehabil* 32: 361–368, 2023.
20. Counts BR, Dankel SJ, Barnett BE, et al. Influence of relative blood flow restriction pressure on muscle activation and muscle adaptation. *Muscle Nerve* 53: 438–445, 2016.
21. de Queiros VS, Rolnick N, Schoenfeld BJ, et al. Hypertrophic effects of low-load blood flow restriction training with different repetition schemes: A systematic review and meta-analysis. *PeerJ* 12: e17195, 2024.
22. Demirtaş B, Çetin O, Çakır E, Beyleroğlu M. The effect of three different sets method used in resistance training on hypertrophy and maximal strength changes. *Phys Edu Students* 26: 270–279, 2022.
23. Drinkwater EJ, Lawton TW, Mckenna MJ, Lindsell RP, Hunt PH, Pyne DB. Increased number of forced repetitions does not enhance strength development with resistance training. *J Strength Cond Res* 21: 841–847, 2007.
24. Enes A, Alves RC, Schoenfeld BJ, et al. Rest-pause and drop-set training elicit similar strength and hypertrophy adaptations compared with

traditional sets in resistance-trained males. *Appl Physiol Nutr Metab* 46: 1417–1424, 2021.
25. Enes A, Oneda G, Leonel DF, et al. Drop-set resistance training versus pyramidal and traditional sets elicits greater psychophysiological responses in men. *Perceptual Mot Skills* 130: 1624–1643, 2023.
26. Escalante G, Gonzalez AM. Building a balanced and symmetrical physique–is regional hypertrophy possible? *Natl Strength Cond Assoc* 9: 6–8, 2022.
27. Evangelista AL, De Souza EO, Moreira DC, et al. Interset stretching vs. traditional strength training: Effects on muscle strength and size in untrained individuals. *J Strength Cond Res* 33: S159–S166, 2019.
28. Faria WFd, Farias JPd, Corrêa RC, et al. Effect of exercise order on the number of repeats and training volume in the tri-set training method. *Rev Bras Cineantropom Desempenho Hum* 18: 187–196, 2016.
29. Farup J, De Paoli F, Bjerg K, Riis S, Ringgard S, Vissing K. Blood flow restricted and traditional resistance training performed to fatigue produce equal muscle hypertrophy. *Scand J Med Sci Sports* 25: 754–763, 2015.
30. Fasihiyan M, Forbes S, Taheri M, et al. Effects of a single or multi-step drop-set training compared to traditional resistance training on muscle performance and body composition. *Scientific J Sport Perform* 2: 410–422, 2023.
31. Fink J, Schoenfeld B, Kikuchi N, Nakazato K. Effects of drop set resistance training on acute stress indicators and long-term muscle hypertrophy and strength. *The J Sports Med Phys Fitness* 58: 597–605, 2018.
32. Fisher JP, Carlson L, Steele J. The effects of breakdown set resistance training on muscular performance and body composition in young men and women. *J Strength Cond Res* 30: 1425–1432, 2016.
33. Fisher JP, Carlson L, Steele J, Smith D. The effects of pre-exhaustion, exercise order, and rest intervals in a full-body resistance training intervention. *Appl Physiol Nutr Metab* 39: 1265–1270, 2014.
34. Fujita RA, Silva NR, Bedo BL, Gomes MM. The pre-exhaustion method does not increase muscle activity in target muscle during strength training in untrained individuals. *J Hum Kinetics* 82: 17–26, 2022.
35. Gantois P, de Souza Fonseca F, de Lima-Júnior D, et al. Acute effects of muscle failure and training system (traditional vs. rest-pause) in resistance exercise on countermovement jump performance in trained adults. *Isokinetics Exerc Sci* 29: 11–20, 2021.
36. Geng Y, Wu X, Zhang Y, Zhang M. Potential moderators of the effects of blood flow restriction training on muscle strength and hypertrophy: A meta-analysis based on a comparison with high-load resistance training. *Sports Med Open* 10: 58, 2024.
37. Gentil P, Oliveira E, Fontana K, Molina G, Oliveira RJd, Bottaro M. Efeitos agudos de vários métodos de treinamento de força no lactato sanguíneo e características de cargas em homens treinados recreacionalmente. *Rev Bras Med Esporte* 12: 303–307, 2006.
38. Gentil P, Oliveira E, Júnior VdAR, do Carmo J, Bottaro M. Effects of exercise order on upper-body muscle activation and exercise performance. *J Strength Cond Res* 21: 1082–1086, 2007.
39. Goto M, Maeda C, Hirayama T, et al. Partial range of motion exercise is effective for facilitating muscle hypertrophy and function through sustained intramuscular hypoxia in young trained men. *J Strength Cond Res* 33: 1286–1294, 2019.
40. Gschneidner D, Carlson L, Steele J, Fisher J. The effects of lengthened-partial range of motion resistance training of the limbs on arm and thigh muscle cross-sectional area: A multi-site cluster trial. *SportRx Preprint*. 2024.
41. Hackett DA, Amirthalingam T. A brief review of forced repetitions for the promotion of muscular hypertrophy. *Strength Cond J* 37: 14–20, 2015.
42. Hackett DA, Johnson NA, Chow C-M. Training practices and ergogenic aids used by male bodybuilders. *J Strength Cond Res* 27: 1609–1617, 2013.
43. Hughes L, Grant I, Patterson SD. Aerobic exercise with blood flow restriction causes local and systemic hypoalgesia and increases circulating opioid and endocannabinoid levels. *J Appl Physiol* 131: 1460–1468, 2021.
44. Hughes L, Patterson SD. The effect of blood flow restriction exercise on exercise-induced hypoalgesia and endogenous opioid and endocannabinoid mechanisms of pain modulation. *J Appl Physiol* 128: 914–924, 2020.
45. Hurley KS, Flippin KJ, Blom LC, Bolin JE, Hoover DL, Judge LW. Practices, perceived benefits, and barriers to resistance training among women enrolled in college. *Int J Exerc Sci* 11: 226–238, 2018.
46. Iversen VM, Eide VB, Unhjem BJ, Fimland MS. Efficacy of supersets versus traditional sets in whole-body multiple-joint resistance training: A randomized controlled trial. *J Strength Cond Res* 38: 1372–1378, 2024.
47. Karimifard M, Arazi H, Mehrabani J. Twelve weeks rest–pause and traditional resistance training: Effects on myokines and performance adaptations among recreationally trained men. *Stresses* 3: 302–315, 2023.
48. Kassiano W, Costa B, Kunevaliki G, et al. Greater gastrocnemius muscle hypertrophy after partial range of motion training performed at long muscle lengths. *J Strength Cond Res* 10: 1519, 2022.
49. Kassiano W, de Vasconcelos Costa BD, Nunes JP, Aguiar AF, de Salles BF, Ribeiro AS. Are we exploring the potential role of specialized techniques in muscle hypertrophy? *Int J Sports Med* 42: 494–496, 2021.
50. Keskin K, Gogus FN, Gunay M, Fujita RA. Equated volume load: Similar improvements in muscle strength, endurance, and hypertrophy for traditional, pre-exhaustion, and drop sets in resistance training. *Sport Sci Health*: 1–10, 2024.
51. Kolind MI, Gam S, Phillip JG, et al. Effects of low load exercise with and without blood-flow restriction on microvascular oxygenation, muscle excitability and perceived pain. *Eur J Sport Sci* 23: 542–551, 2023.
52. Larsen S, Sandvik Kristiansen B, Swinton PA, et al. The effects of hip flexion angle on quadriceps femoris muscle hypertrophy in the leg extension exercise. *J Sports Sci* 43: 210–221, 2024.
53. Larsen S, Swinton P, Sandberg NØ, et al. Resistance training beyond momentary failure: The effects of past-failure partials on muscle hypertrophy in the gastrocnemius. *Front Psychol* 16: 1494323, 2025.
54. Lasevicius T, Ugrinowitsch C, Schoenfeld BJ, et al. Effects of different intensities of resistance training with equated volume load on muscle strength and hypertrophy. *Eur J Sport Sci* 18: 772–780, 2018.
55. Maeo S, Huang M, Wu Y, et al. Greater hamstrings muscle hypertrophy but similar damage protection after training at long versus short muscle lengths. *Med Sci Sports Exerc* 53: 825–837, 2020.
56. Maeo S, Wu Y, Huang M, et al. Triceps brachii hypertrophy is substantially greater after elbow extension training performed in the overhead versus neutral arm position. *Eur J Sport Sci* 23: 1240–1250, 2023.
57. Mang Z, Beam J, Vigil E, Martinez A. The acute effect of ascending-pyramid, descending-pyramid, and constant-load set configurations on repetition performance, training volume, and barbell velocity during bench press exercise. *Int J Strength Cond*. 2024;4:1–15.
58. Marshall PW, Robbins DA, Wrightson AW, Siegler JC. Acute neuromuscular and fatigue responses to the rest-pause method. *J Sci Med Sport* 15: 153–158, 2012.
59. McMahon G, Morse CI, Burden A, Winwood K, Onambélé GL. Muscular adaptations and insulin-like growth factor-1 responses to resistance training are stretch-mediated. *Muscle Nerve* 49: 108–119, 2014.
60. McMahon GE, Morse CI, Burden A, Winwood K, Onambélé GL. Impact of range of motion during ecologically valid resistance training protocols on muscle size, subcutaneous fat, and strength. *J Strength Cond Res* 28: 245–255, 2014.
61. Miranda H, Figueiredo T, Rodrigues B, Paz GA, Simão R. Influence of exercise order on repetition performance among all possible combinations on resistance training. *Res Sports Med* 21: 355–366, 2013.
62. Nascimento DdC, Rolnick N, Neto IVdS, Severin R, Beal FLR. A useful blood flow restriction training risk stratification for exercise and rehabilitation. *Front Physiol* 13: 808622, 2022.
63. Nunes JP, Blazevich AJ, Schoenfeld BJ, et al. Determining changes in muscle size and architecture after exercise training: One site does not fit all. *J Strength Cond Res* 38: 787–790, 2024.
64. Nunes JP, Grgic J, Cunha PM, et al. What influence does resistance exercise order have on muscular strength gains and muscle hypertrophy? A systematic review and meta-analysis. *Eur J Sport Sci* 21: 149–157, 2021.
65. Nunes JP, Jacinto JL, Ribeiro AS, et al. Placing greater torque at shorter or longer muscle lengths? Effects of cable vs. barbell preacher curl training on muscular strength and hypertrophy in young adults. *Int J Environ Res Public Health* 17: 5859, 2020.
66. Oranchuk DJ, Storey AG, Nelson AR, Cronin JB. Isometric training and long-term adaptations: Effects of muscle length, intensity, and intent: A systematic review. *Scand J Med Sci Sports* 29: 484–503, 2019.
67. Ozaki H, Kubota A, Natsume T, et al. Effects of drop sets with resistance training on increases in muscle CSA, strength, and endurance: A pilot study. *J Sports Sci* 36: 691–696, 2018.
68. Pallarés JG, Hernández-Belmonte A, Martínez-Cava A, Vetrovsky T, Steffl M, Courel-Ibáñez J. Effects of range of motion on resistance training adaptations: A systematic review and meta-analysis. *Scand J Med Sci Sports* 31: 1866–1881, 2021.
69. Patterson SD, Hughes L, Warmington S, et al. Blood flow restriction exercise: Considerations of methodology, application, and safety. *Front Physiol* 10: 533, 2019.
70. Paunksnis MR, Evangelista AL, La Scala Teixeira CV, et al. Metabolic and hormonal responses to different resistance training systems in elderly men. *The Aging Male* 21: 106–110, 2018.
71. Pedrosa GF, Lima FV, Schoenfeld BJ, et al. Partial range of motion training elicits favorable improvements in muscular adaptations when carried out at long muscle lengths. *Eur J Sport Sci* 22: 1250–1260, 2022.
72. Pedrosa GF, Simões MG, Figueiredo MO, et al. Training in the initial range of motion promotes greater muscle adaptations than at Final in the arm curl. *Sports* 11: 39, 2023.
73. Pinto RS, Gomes N, Radaelli R, Botton CE, Brown LE, Bottaro M. Effect of range of motion on muscle strength and thickness. *J Strength Cond Res* 26: 2140–2145, 2012.
74. Prestes J, Tibana RA, de Araujo Sousa E, et al. Strength and muscular adaptations after 6 weeks of rest-pause vs. traditional multiple-sets

resistance training in trained subjects. *J Strength Cond Res* 33: S113–S121, 2019.

75. Ribeiro AS, Aguiar AF, Schoenfeld BJ, et al. Effects of different resistance training systems on muscular strength and hypertrophy in resistance-trained older women. *J Strength Cond Res* 32: 545–553, 2018.
76. Ribeiro AS, Nunes JP, Cunha PM, Aguiar AF, Schoenfeld BJ. Potential role of pre-exhaustion training in maximizing muscle hypertrophy: A review of the literature. *Strength Cond J* 41: 75–80, 2019.
77. Ribeiro AS, Schoenfeld BJ, Fleck SJ, Pina FLC, Nascimento MA, Cyrino ES. Effects of traditional and pyramidal resistance training systems on muscular strength, muscle mass, and hormonal responses in older women: A randomized crossover trial. *J Strength Cond Res* 31: 1888–1896, 2017.
78. Ribeiro AS, Silva DRPd, Nascimento MAd, Avelar A, Ritti-Dias RM, Cyrino ES. Effect of the manipulation of exercise order in the tri-set training system. *Revista Brasileira de Cineantropometria & Desempenho Humano* 15: 527–534, 2013.
79. Robbins DW, Young WB, Behm DG. The effect of an upper-body agonist-antagonist resistance training protocol on volume load and efficiency. *J Strength Cond Res* 24: 2632–2640, 2010.
80. Rolnick N. Unpacking the blood flow restriction device features literature: Multi-chambered bladder design. *Front Sports Act Living* 6: 1457539, 2024.
81. Rolnick N, de Queiros VS, Moghaddam M, et al. Cardiovascular, perceptual, and performance responses to single-vs. multi-chambered blood flow restriction cuffs. *Front Sports Act Living* 6: 1469356, 2024.
82. Rolnick N, Kimbrell K, Cerqueira MS, Weatherford B, Brandner C. Perceived barriers to blood flow restriction training. *Front Rehabil Sci* 2: 697082, 2021.
83. Rolnick N, Kimbrell K, de Queiros V. Beneath the cuff: Often overlooked and under-reported blood flow restriction device features and their potential impact on practice—a review of the current state of the research. *Front Physiol* 14: 1089065, 2023.
84. Rolnick N, Schoenfeld BJ. Blood flow restriction training and the physique athlete: A practical research-based guide to maximizing muscle size. *Strength Cond J* 42: 22–36, 2020.
85. Sato S, Yoshida R, Kiyono R, et al. Elbow joint angles in elbow flexor unilateral resistance exercise training determine its effects on muscle strength and thickness of trained and non-trained arms. *Front Physiol* 12: 734509, 2021.
86. Schoenfeld BJ, Grgic J, Ogborn D, Krieger JW. Strength and hypertrophy adaptations between low-vs. High-load resistance training: A systematic review and meta-analysis. *J Strength Cond Res* 31: 3508–3523, 2017.
87. Schoenfeld BJ, Wackerhage H, De Souza E. Inter-set stretch: A potential time-efficient strategy for enhancing skeletal muscle adaptations. *Front Sports Act Living* 4: 1035190, 2022.
88. Siewe J, Marx G, Knöll P, et al. Injuries and overuse syndromes in competitive and elite bodybuilding. *Int J Sports Med* 35: 943–948, 2014.
89. Soares EG, Brown LE, Gomes WA, et al. Comparison between pre-exhaustion and traditional exercise order on muscle activation and performance in trained men. *J Sports Sci Med* 15: 111–117, 2016.
90. Sødal LK, Kristiansen E, Larsen S, van den Tillaar R. Effects of drop sets on skeletal muscle hypertrophy: A systematic review and meta-analysis. *Sports Med Open* 9: 66, 2023.
91. Song JS, Spitz RW, Yamada Y, et al. Exercise-induced hypoalgesia and pain reduction following blood flow restriction: A brief review. *Phys Ther Sport* 50: 89–96, 2021.
92. Trindade TB, Alves RC, de Castro BM, et al. Pre-exhaustion training, a narrative review of the acute responses and chronic adaptations. *Int J Exerc Sci* 15: 507–525, 2022.
93. Trindade TB, Prestes J, Neto LO, et al. Effects of pre-exhaustion versus traditional resistance training on training volume, maximal strength, and quadriceps hypertrophy. *Front Physiol* 10: 1424, 2019.
94. Valamatos MJ, Tavares F, Santos RM, Veloso AP, Mil-Homens P. Influence of full range of motion vs. equalized partial range of motion training on muscle architecture and mechanical properties. *Eur J Appl Physiol* 118: 1969–1983, 2018.
95. Van Every DW, Coleman M, Rosa A, et al. Loaded inter-set stretch may selectively enhance muscular adaptations of the plantar flexors. *PLoS One* 17: e0273451, 2022.
96. Varovic D, Wolf M, Steele J, Schoenfeld BJ, Grgic J, Mikulic P. Regional hypertrophy with resistance training—does muscle length matter? A systematic review and meta-analysis. *SportRx Preprint*. 2024.
97. Varović D, Žganjer K, Vuk S, Schoenfeld BJ. Drop-set training elicits differential increases in non-uniform hypertrophy of the quadriceps in leg extension exercise. *Sports* 9: 119, 2021.
98. Vilaça-Alves J, Geraldes L, Fernandes HM, et al. Effects of pre-exhausting the biceps brachii muscle on the performance of the front lat pull-down exercise using different handgrip positions. *J Hum Kinetics* 42: 157–163, 2014.
99. Wadhi T, Barakat C, Evangelista AL, et al. Loaded inter-set stretching for muscular adaptations in trained males: Is the hype real? *Int J Sports Med* 43: 168–176, 2022.
100. Wallace W, Ugrinowitsch C, Stefan M, et al. Repeated bouts of advanced strength training techniques: Effects on volume load, metabolic responses, and muscle activation in trained individuals. *Sports* 7: 14, 2019.
101. Werkhausen AE, E Solberg C, Paulsen G, Bojsen-Møller J, Seynnes OR. Adaptations to explosive resistance training with partial range of motion are not inferior to full range of motion. *Scand J Med Sci Sports* 31: 1026–1035, 2021.
102. Wolf M, Androulakis-Korakakis P, Fisher J, Schoenfeld B, Steele J. Partial vs full range of motion resistance training: A systematic review and meta-analysis. *Int J Strength Cond*. 2023;3: 1–21.
103. Wolf M, Androulakis Korakakis P, Piñero A, et al. Lengthened partial repetitions elicit similar muscular adaptations as full range of motion repetitions during resistance training in trained individuals. *PeerJ* 13: e18904, 2025.
104. Yang J, Rolnick N, Merriwether E, Rao S. Hypoalgesia and conditioned pain modulation in blood flow restriction resistance exercise. *Int J Sports Med* 45: 810–819, 2024.
105. Zabaleta-Korta A, Fernández-Peña E, Torres-Unda J, Francés M, Zubillaga A, Santos-Concejero J. Regional hypertrophy: The effect of exercises at long and short muscle lengths in recreationally trained women. *J Hum Kinetics* 88: 259, 2023.
106. Zhang X, Weakley J, Li H, Li Z, García-Ramos A. Superset versus traditional resistance training prescriptions: A systematic review and meta-analysis exploring acute and chronic effects on mechanical, metabolic, and perceptual variables. *Sports Med* 55: 953–975, 2025.